

Product Line Profitability and Margin Performance Analysis

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Abstract—For distribution businesses, high sales volume does not necessarily translate into strong profitability. Products with significant revenue contribution may generate low margins, while certain low-volume products may contribute disproportionately to overall profit. This study presents a comprehensive Product Line Profitability and Margin Performance Analysis for Nassau Candy Distributor using transactional sales data containing over 10,000 records.

The analysis focuses on evaluating product profitability, division-level financial performance, profit concentration, and cost structure efficiency. A structured analytical framework involving data cleaning, profitability metric calculation, exploratory data analysis, Pareto analysis, and cost diagnostics was developed to identify financially efficient and inefficient products.

Results indicate that a small group of chocolate products contributes the majority of both revenue and profit, while several low-performing products exhibit margin-risk characteristics due to unfavorable cost structures. Division-level analysis reveals significant differences in financial efficiency, highlighting areas requiring pricing and cost optimization.

An interactive Streamlit dashboard was developed to enable dynamic exploration of profitability metrics through filters, search capabilities, and visual analytics. The findings demonstrate how data-driven profitability analysis can support strategic pricing decisions, product portfolio optimization, and long-term financial sustainability.

Index Terms—Profitability Analysis, Gross Margin, Product Performance, Cost Optimization, Business Intelligence, Streamlit Dashboard, Data Analytics.

I. INTRODUCTION

In highly competitive distribution environments, revenue growth alone is not a sufficient indicator of business success. Organizations must understand the profitability generated by individual products, product divisions, and customer demand patterns to ensure sustainable financial performance. Products with strong sales volumes may appear successful on the surface while generating relatively low margins due to elevated manufacturing, sourcing, or operational costs.

Nassau Candy Distributor manages a diverse portfolio of confectionery products across multiple product divisions. As product portfolios expand, management requires greater visibility into which products truly drive profitability and which products introduce margin risk. Without such visibility, decisions related to pricing, promotions, inventory planning, and product rationalization become reactive rather than strategically informed.

This project introduces a data-driven framework for evaluating product profitability and margin performance. By com-

bing financial metrics, exploratory data analysis, Pareto-based profit concentration analysis, and cost diagnostics, the study provides actionable insights into product performance and division-level efficiency.

Furthermore, an interactive Streamlit dashboard was developed to support business users in monitoring profitability trends and identifying improvement opportunities through real-time analytical exploration.

II. PROBLEM STATEMENT

Despite maintaining extensive transactional sales records, Nassau Candy Distributor lacks a structured analytical framework for evaluating product profitability and margin performance. Management currently has limited visibility into several critical business questions:

1. Which product lines generate the highest gross margins.
2. Whether high-sales products are also the most profitable products.
3. How profitability differs across product divisions.
4. Which products contribute most significantly to total profit.
5. Which products exhibit margin-risk characteristics due to unfavorable cost structures.
6. Whether revenue generation is excessively dependent on a small number of products.

Without these insights, strategic decisions regarding pricing optimization, product expansion, sourcing negotiations, and portfolio rationalization remain largely intuition-driven. This creates the risk of allocating resources toward products that contribute revenue but fail to maximize profitability.

The primary objective of this study is therefore to transform raw transactional sales data into actionable financial intelligence capable of supporting data-driven business decision-making.

III. OBJECTIVES

The primary objectives of this project are:

- 1) To calculate and evaluate key profitability metrics for individual products.
- 2) To identify products with the highest profit contribution and gross margin performance.
- 3) To compare financial efficiency across product divisions.
- 4) To assess revenue and profit concentration using Pareto analysis.

- 5) To identify margin-risk products through cost structure diagnostics.
- 6) To provide actionable recommendations regarding pricing, sourcing, and product portfolio management.
- 7) To develop an interactive Streamlit dashboard for profitability monitoring and decision support.

IV. SYSTEM ARCHITECTURE

The analytical framework developed for this project follows a structured business intelligence pipeline designed to transform raw sales transactions into actionable profitability insights.

The pipeline consists of the following stages:

- 1) Data Collection – Loading transactional sales records.
- 2) Data Cleaning – Validation of sales, cost, and profit information.
- 3) KPI Calculation – Generation of profitability metrics.
- 4) Exploratory Data Analysis – Statistical and visual analysis of sales and profit behavior.
- 5) Product Profitability Analysis – Product ranking and classification.
- 6) Division Performance Analysis – Division-level financial evaluation.
- 7) Pareto Analysis – Revenue and profit concentration assessment.
- 8) Cost Diagnostics – Margin risk and pricing efficiency evaluation.
- 9) Dashboard Development – Interactive visualization using Streamlit.

V. DATASETS DESCRIPTION

The dataset consists of 10,194 transactional sales records representing product sales across multiple product divisions and geographic regions.

The dataset contains operational, financial, and geographic information required for profitability evaluation.

Key attributes include:

- Order Date
- Product Name
- Division
- Region
- State/Province
- Sales
- Units
- Gross Profit
- Cost

The product portfolio is categorized into three primary divisions:

- Chocolate
- Sugar
- Other

These attributes enable profitability analysis at both the transaction level and aggregated business levels such as product, division, region, and state.

VI. METHODOLOGY

The project follows a structured data analysis approach:

- 1) **Data Cleaning and Validation:** The first stage of the analysis involved cleaning and validating the raw transactional dataset to ensure data quality and analytical reliability. Since profitability calculations are highly sensitive to inaccuracies in sales, cost, and profit values, several preprocessing steps were performed before conducting any business analysis. The following validation procedures were implemented:
 - Verification of sales and cost values to ensure numerical consistency.
 - Identification and removal of records containing invalid sales values or inconsistent profit calculations.
 - Detection and treatment of missing unit quantities.
 - Standardization of product names and division labels to eliminate duplicate categories caused by formatting inconsistencies.
 - Conversion of date fields into a standardized datetime format to support temporal analysis.
 - Validation of gross profit values to ensure alignment with corresponding sales and cost information.
 These preprocessing steps ensured that all subsequent profitability calculations were based on accurate and reliable data.
- 2) **Profitability Metric Calculation:** Following data preparation, a set of key profitability indicators was calculated to evaluate product-level financial performance. The metrics were selected to provide visibility into profitability, margin efficiency, and contribution to overall business performance. The following Key Performance Indicators (KPIs) were computed:
 1. Gross Margin (%)

Gross Margin represents the percentage of revenue retained after accounting for product cost.

$$\text{Gross Margin (\%)} = (\text{Gross Profit} / \text{Sales}) \times 100$$
 2. Profit per Unit

This metric measures the amount of profit generated for every unit sold.

$$\text{Profit per Unit} = \text{Gross Profit} / \text{Units}$$
 3. Revenue Contribution

Revenue Contribution quantifies the share of total company revenue generated by each product.

$$\text{Revenue Contribution} = \text{Product Sales} / \text{Total Sales}$$
 4. Profit Contribution

Profit Contribution measures the percentage of total company profit generated by each product.

$$\text{Profit Contribution} = \text{Product Profit} / \text{Total Profit}$$

These indicators formed the foundation for product ranking, division comparison, and concentration analysis.
- 3) **Product-Level Profitability Analysis:** Product-level profitability analysis was conducted to determine which products generate the greatest financial value for the organization.

The analysis involved aggregating sales, profit, cost, and unit information at the product level. Products were subsequently ranked according to total gross profit and average gross margin.

To facilitate business interpretation, products were classified into profitability categories based on sales and margin performance:

- High Sales & High Margin
- High Sales & Low Margin
- Low Sales but High Margin
- Low Sales & Low Profit

This classification enabled the identification of products suitable for expansion, products requiring margin improvement, and products potentially requiring discontinuation review.

The analysis further highlighted the relationship between sales performance and actual profitability, ensuring that revenue-driven decisions could be evaluated alongside financial efficiency considerations.

- 4) **Division-Level Performance Analysis:** To evaluate organizational performance at a broader business level, financial metrics were aggregated across product divisions.

The objective of this analysis was to compare divisions based on revenue generation, profit contribution, margin efficiency, and operational effectiveness.

For each division, the following metrics were calculated:

- Total Sales
- Total Gross Profit
- Total Units Sold
- Average Gross Margin
- Profit per Unit

Comparative analysis was then performed to identify divisions exhibiting strong financial efficiency and divisions displaying structural profitability challenges.

Based on profitability characteristics, divisions were categorized into performance groups such as:

- High Efficiency
- Structural Margin Issue
- Efficient but Low Scale

This approach enabled the identification of strategic opportunities for resource allocation, pricing optimization, and portfolio expansion.

- 5) **Profit Concentration (Pareto) Analysis;** The Pareto Principle, commonly referred to as the 80/20 Rule, was applied to evaluate the concentration of revenue and profit generation across products.

Products were ranked in descending order of revenue and profit contribution. Cumulative percentages were then calculated to determine the minimum number of products responsible for approximately 80% of total business performance.

The objectives of the Pareto analysis were:

- Identify products contributing to 80% of total revenue.
- Identify products contributing to 80% of total profit.
- Evaluate dependency on a small number of products.

- Detect potential concentration risks.

In addition to product concentration, regional revenue distribution was analyzed to assess geographic dependency patterns and identify high-performing regions.

This analysis provided management with a clearer understanding of business concentration and portfolio risk exposure.

- 6) **Cost Structure Diagnostics** Cost Structure Diagnostics were performed to evaluate the relationship between product cost and financial performance.

The primary objective was to identify products that consume significant cost resources while generating relatively low profitability.

The analysis included:

- Cost versus Sales relationship assessment.
- Margin evaluation across products.
- Detection of cost-heavy and margin-poor products.
- Identification of potential pricing inefficiencies.

Products exhibiting high cost ratios combined with weak margins were classified as margin-risk products.

Based on the diagnostic results, products were assigned business recommendations including:

- Maintain and Scale
- Repricing Review
- Cost Renegotiation
- Discontinuation Review

These recommendations provide a structured framework for improving overall profitability and optimizing resource allocation.

- 7) **Exploratory Data Analysis (EDA):** Exploratory Data Analysis (EDA) was conducted to understand underlying data patterns, profitability behavior, revenue distribution, cost dynamics, and regional performance.

More than fifteen visualizations were developed to examine relationships among key variables including sales, cost, profit, units sold, divisions, products, and geographic regions.

The EDA process focused on:

- Distribution analysis
- Product performance analysis
- Division performance comparison
- Correlation analysis
- Time-series analysis
- Regional profitability analysis

The insights derived from EDA provided the foundation for subsequent profitability diagnostics and strategic recommendations.

- 8) **Dashboard Development:** To facilitate business decision-making and improve accessibility of analytical findings, an interactive dashboard was developed using Streamlit.

The dashboard integrates all major analytical outputs into a unified business intelligence platform.

Key dashboard modules include:

- Product Profitability Overview
- Division Performance Dashboard

- Cost vs Margin Diagnostics
 - Profit Concentration Analysis
- Interactive user controls include:
- Date Range Selection
 - Division Filtering
 - Margin Threshold Adjustment
 - Product Search Functionality

The dashboard enables stakeholders to dynamically explore profitability trends and investigate product performance without requiring technical expertise.

VII. CONSIDERATION OF MACHINE LEARNING TECHNIQUES

During the project design phase, clustering-based techniques such as K-Means were evaluated for potential implementation. However, the primary objective of this study was profitability assessment rather than customer segmentation or market grouping.

Since the dataset was primarily structured around transactional financial metrics and the business requirements focused on margin analysis, profit contribution, division performance, and cost diagnostics, traditional business analytics techniques provided more interpretable and actionable insights than unsupervised clustering methods. Consequently, the study prioritized profitability metrics, Pareto analysis, and financial performance diagnostics over machine learning-based segmentation approaches. Future research may explore clustering techniques for customer segmentation, demand pattern analysis, or product grouping strategies.

VIII. DATA ANALYSIS AND VISUALIZATION

This section presents the exploratory and profitability-focused analysis conducted on the Nassau Candy Distributor dataset. The objective was to identify profitable products, understand margin behavior, evaluate divisional performance, and uncover revenue-profit relationships that support strategic business decisions. Unlike customer segmentation projects, this study focuses on financial performance measurement. Therefore, clustering techniques such as K-Means were not applied because the dataset primarily contains transactional sales, cost, profit, and product information rather than customer behavioral attributes suitable for segmentation. Instead, profitability metrics, aggregation analysis, correlation analysis, and Pareto-based evaluation were utilized to derive actionable insights.

- Cost vs Sales Analysis:** The Cost vs Sales scatter plot illustrates the relationship between product sales revenue and associated manufacturing cost. A strong positive relationship is observed, indicating that products generating higher sales generally incur proportionally higher production costs. However, several products demonstrate significantly higher sales relative to their costs, suggesting superior profitability and pricing efficiency.

These products represent strategic assets within the portfolio because they contribute substantial revenue while maintaining healthy profit margins. The visualization also helps identify products with disproportionately high costs that may require supplier renegotiation, production optimization, or pricing adjustments. Such products represent potential margin risks if cost growth exceeds revenue growth.

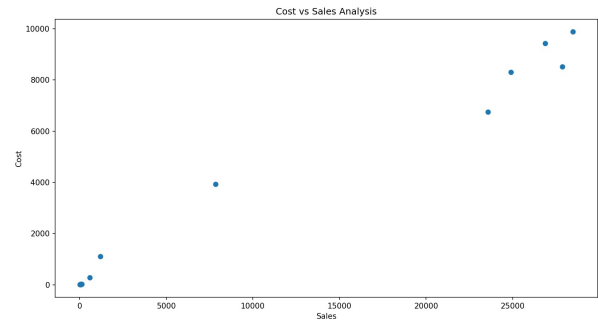


Fig. 1. Cost vs Sales Analysis

- Sales Distribution:** The sales distribution histogram reveals a highly right-skewed pattern. Most transactions occur within relatively low sales ranges, while a limited number of transactions contribute exceptionally high sales values. This pattern indicates that Nassau Candy's revenue generation is concentrated among a small subset of high-performing orders and products. Such concentration is common in distribution businesses where a limited number of products account for a significant share of overall revenue. The long-tail behavior observed in the distribution suggests opportunities for product portfolio optimization and inventory prioritization.

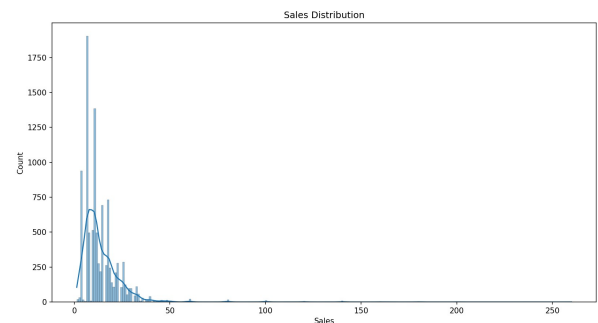


Fig. 2. Sales Distribution

- Gross Profit Distribution:** The gross profit distribution follows a similar right-skewed pattern as sales distribution. Most orders generate modest profit values, while only a small number of transactions contribute exceptionally high profits.

This finding confirms that profitability is concentrated among selected products rather than being evenly distributed across the product catalog. The analysis highlights the importance of identifying and protecting high-profit products through strategic pricing, promotional support, and inventory planning.

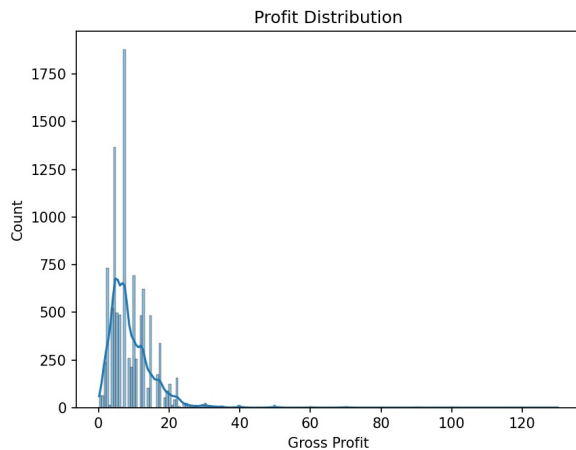


Fig. 3. Profit Distribution

- d) **Units Distribution:** The units distribution demonstrates that most customer orders involve relatively small quantities, typically between two and six units.

Higher unit purchases occur less frequently and represent a smaller proportion of total transactions. This indicates that customer demand is distributed across numerous smaller orders rather than bulk purchases.

Understanding order quantity behavior assists inventory managers in forecasting demand and maintaining optimal stock levels while reducing carrying costs.

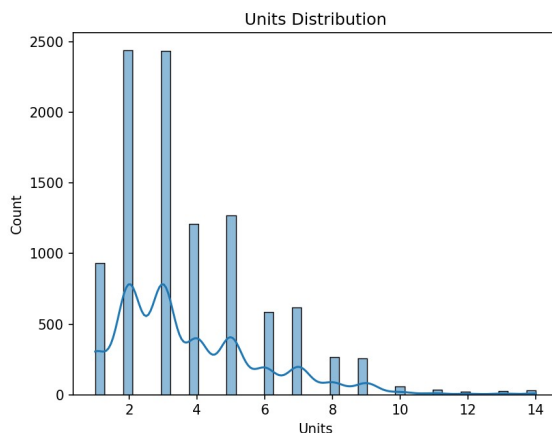


Fig. 4. Units Distribution

- e) **Gross Margin Distribution:** Gross margin percentage represents one of the most important profitability indicators in this study. The distribution shows that the majority of products operate within a margin range of approximately 65% to 72%. This relatively concentrated margin range indicates strong pricing consistency across the portfolio. Only a small number of products fall below 50% margin, suggesting potential pricing inefficiencies or elevated production costs. Products with unusually low margins should be reviewed for cost optimization, supplier negotiations, or possible product rationalization.

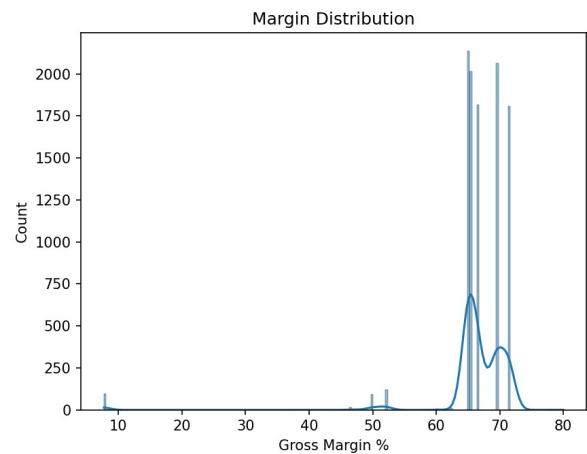


Fig. 5. Margin Distribution

- f) **Top Products by Sales:** The analysis identified several Wonka chocolate products as dominant revenue generators.

Among all products, **Wonka Bar – Triple Dazzle Caramel, Wonka Bar – Scrumdiddlyumptious, and Wonka Bar – Milk Chocolate** generated the highest total sales values.

The concentration of revenue among chocolate products demonstrates the strategic importance of the chocolate division within Nassau Candy's overall business performance.

These products should receive continued operational focus because they drive a significant portion of total revenue generation. Supplier negotiations, or possible product rationalization.

- g) **Top Products by Gross Profit:** Profitability analysis reveals that high sales products are not always the most profitable products.

Although chocolate products continue to dominate profit generation, the ranking differs slightly from the sales ranking. This confirms the necessity of evaluating both revenue and profit simultaneously when making business decisions.

Products such as **Wonka Bar – Scrumdiddlyump-**

tious and Wonka Bar – Triple Dazzle Caramel demonstrate strong financial performance by generating both substantial revenue and significant profit contributions.

These products represent the most valuable components of the product portfolio.

- h) **Sales by Division:** Division-level analysis indicates that the Chocolate Division overwhelmingly dominates total revenue generation.

The chocolate category contributes substantially more sales than both the Sugar and Other divisions combined.

This result suggests that Nassau Candy's revenue stream is heavily dependent on chocolate products. While this concentration creates a strong revenue foundation, it also introduces dependency risk should demand conditions change within that product category.

Diversification efforts may therefore be beneficial to reduce long-term business risk.

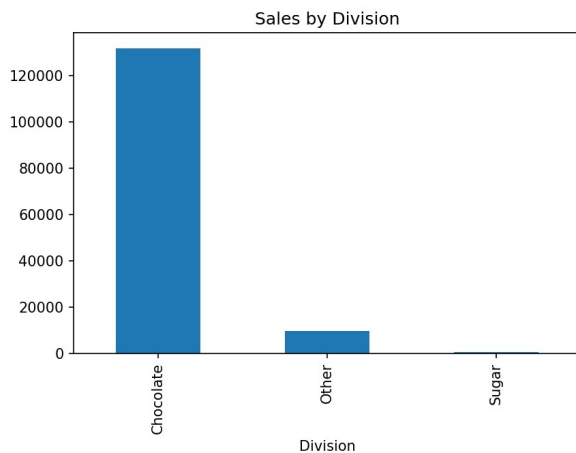


Fig. 6. Sales by Division

- i) **Average Margin by Division:** While the Chocolate Division generated the highest revenue, it also maintained the strongest average margin performance, exceeding both the Sugar and Other divisions.

The Other division exhibited the lowest average margin percentage, indicating comparatively weaker financial efficiency.

This finding suggests that certain products within the Other category may require pricing adjustments, cost reductions, or strategic review to improve profitability.

Division-level margin analysis provides management with valuable insight beyond revenue-based performance evaluation.

- j) **Sales vs Profit Relationship:** A strong positive relationship exists between sales and gross profit.

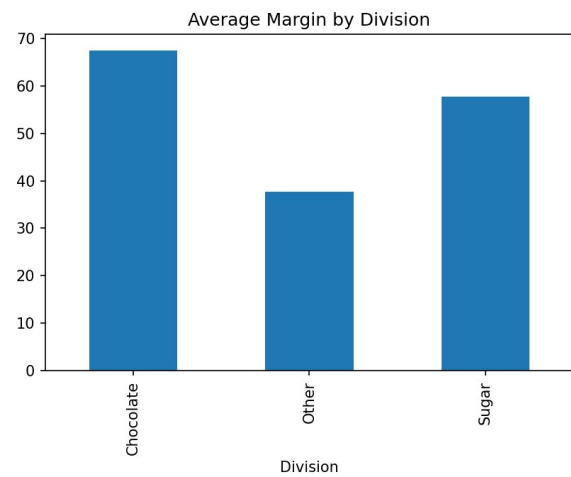


Fig. 7. Average Margin by Division

As sales increase, profit generally increases at a proportional rate.

The pattern indicates effective pricing practices and stable profit generation across most products.

However, the presence of some products with relatively high sales but comparatively lower profits suggests potential margin inefficiencies. Such products warrant further investigation to determine whether excessive costs are reducing profitability.

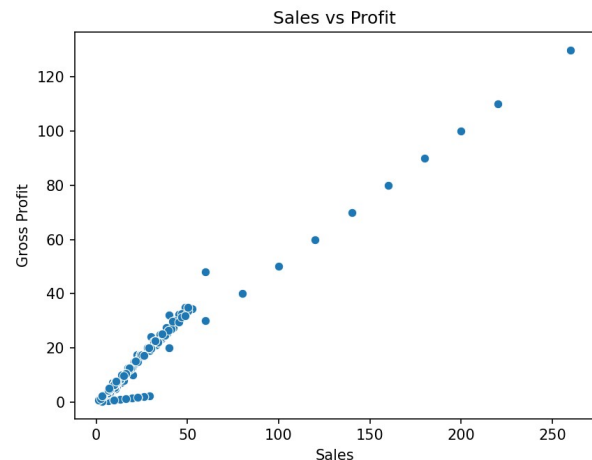


Fig. 8. Sales vs Profit Scatter Plot

- k) **Sales vs Cost Relationship:** The scatter plot demonstrates a direct relationship between sales and production cost. Products generating larger revenue typically require higher manufacturing expenditures.

Despite this relationship, several products achieve stronger sales growth relative to cost increases, indicating better operational efficiency.

This analysis assists management in identifying products that maximize revenue generation without

proportionately increasing production costs.

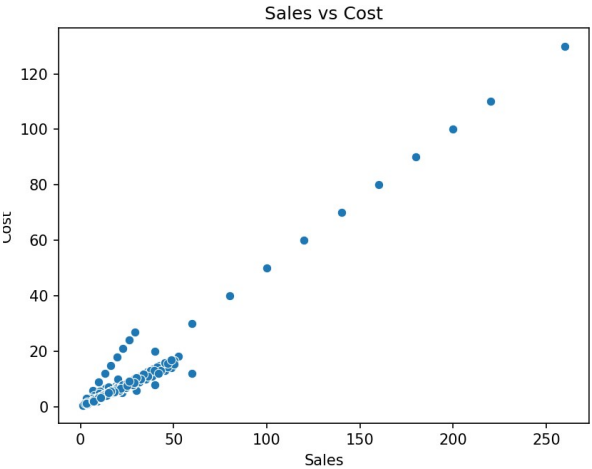


Fig. 9. Sales vs Cost Relationship Scatter Plot

- l) **Correlation Analysis**The correlation matrix provides a quantitative assessment of relationships among key business metrics. The strongest correlation was observed between:
- Sales and Gross Profit (0.98)
 - Sales and Cost (0.96)
- These values indicate that revenue growth strongly influences profitability while simultaneously driving production costs. Units sold also exhibit positive relationships with both sales and profit, although the correlations are comparatively weaker. The analysis confirms that sales performance remains the primary driver of profitability across Nassau Candy’s product portfolio.



Fig. 10. Correlation Analysis

- m) **Sales Trend Over Time:** The sales trend analysis reveals noticeable fluctuations throughout the

observed period. Although short-term volatility exists, the overall pattern indicates growth in sales activity over time. Several high-revenue peaks are observed, particularly during later periods of the dataset. These peaks may reflect seasonal demand patterns, promotional campaigns, or product-specific sales surges. Time-series analysis enables management to identify periods of strong performance and support future forecasting initiatives.

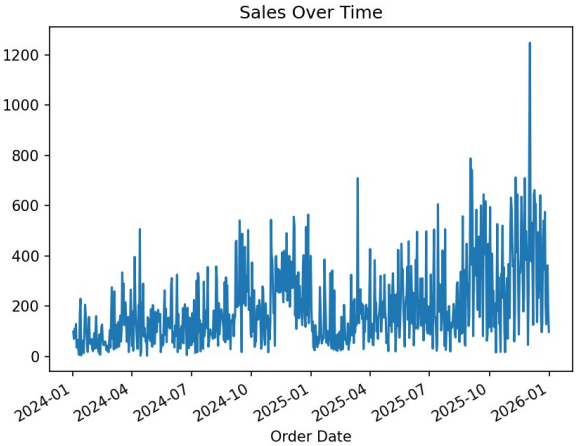


Fig. 11. Sales Trend Over Time

- n) **Profit by Region:** Regional profitability analysis shows that the Pacific Region generated the highest overall profit contribution, followed by the Atlantic and Interior regions. The Gulf Region recorded comparatively lower profitability levels. These findings indicate geographical differences in market performance and may guide future sales expansion strategies, promotional efforts, and distribution planning. Regions demonstrating stronger profit generation should receive continued investment and operational support.

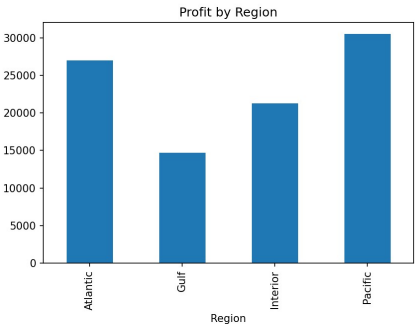


Fig. 12. Profit by Region

- o) **Top States by Sales:** State-level analysis reveals that California generated the highest sales volume among all states, significantly outperforming other regions.

New York and Texas also contributed substantial revenue, forming a group of strategically important markets for Nassau Candy.

The concentration of sales within a limited number of states suggests opportunities for focused marketing initiatives and customer retention programs within these high-performing territories.

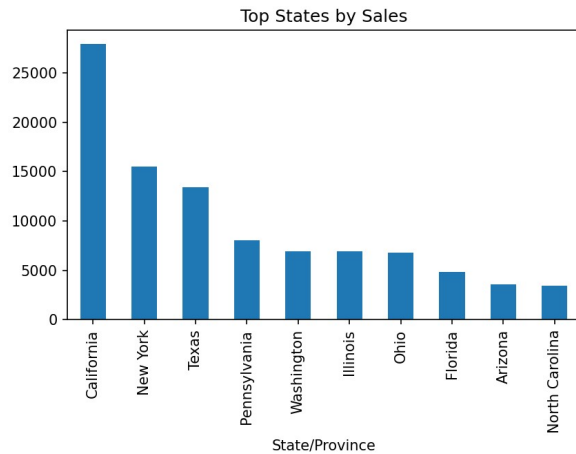


Fig. 13. Top States by Sales

- p) **Section Summary:** The visual analysis demonstrates that profitability within Nassau Candy Distributor is highly concentrated among a small group of chocolate products. The Chocolate Division dominates both revenue and profit generation while maintaining the strongest margin performance. Correlation analysis confirms that sales growth remains the primary driver of profitability, while regional and state-level analyses reveal geographic opportunities for strategic expansion. Collectively, these findings provide a strong foundation for pricing optimization, portfolio rationalization, and profit-focused decision-making.

IX. BUSINESS INSIGHTS

The profitability and margin performance analysis of Nassau Candy Distributor revealed several important business insights regarding product performance, divisional efficiency, cost structures, and geographical profitability patterns.

A. Chocolate Division Dominates Business Performance

The Chocolate Division emerged as the strongest contributor to overall business success. It generated the highest revenue among all divisions while simultaneously maintaining the highest average gross margin percentage.

This indicates that chocolate products are not only popular among customers but are also financially efficient. The division demonstrates a balanced combination of strong demand, effective pricing, and controlled production costs.

As a result, chocolate products represent the primary drivers of both revenue growth and profit generation within Nassau Candy Distributor.

B. Revenue Does Not Always Indicate Profitability

One of the most significant findings of the analysis is that high sales volume does not automatically translate into superior profitability.

Several products generated substantial sales revenue but contributed comparatively lower profits due to higher production costs or lower margin percentages.

This finding validates the need for profitability-focused decision making rather than relying solely on sales figures when evaluating product success.

Organizations that focus only on revenue risk allocating resources toward products that contribute limited financial value.

C. Profit Generation Is Concentrated Among a Small Number of Products

The top-performing Wonka chocolate products account for a disproportionately large share of total profit generation.

Products such as:

1. Wonka Bar – Scrumdiddlyumptious
2. Wonka Bar – Triple Dazzle Caramel
3. Wonka Bar – Milk Chocolate
4. Wonka Bar – Nutty Crunch Surprise

consistently ranked among the highest contributors to both sales and gross profit.

This concentration suggests that Nassau Candy's financial performance depends heavily on a limited subset of products.

While these products create strong profitability, overdependence on a small number of items introduces portfolio risk and should be monitored carefully.

D. Margin Stability Across Most Products

The gross margin distribution demonstrated that most products operate within a relatively narrow margin range of approximately 65%–72%.

This consistency indicates:

1. Effective pricing strategies
2. Stable manufacturing costs
3. Controlled operational performance

Only a small number of products exhibited significantly lower margins, indicating potential inefficiencies that require management attention.

E. Strong Relationship Between Sales and Profit

Correlation analysis showed an extremely strong positive relationship between Sales and Gross Profit.

This finding confirms that increasing sales generally contributes directly to higher profitability.

However, cost analysis revealed that revenue growth

must still be monitored carefully because increasing sales also increases manufacturing costs.

Therefore, sustainable growth should focus on products that maintain strong margins while scaling revenue.

F. Geographic Profitability Differences Exist

Regional analysis revealed meaningful differences in profit contribution across markets.

The Pacific region generated the highest overall profit, followed by the Atlantic region.

Meanwhile, the Gulf region demonstrated comparatively lower profitability.

These variations suggest that regional market conditions, customer preferences, pricing dynamics, or distribution efficiencies influence financial performance.

Understanding these differences can support future expansion and resource allocation decisions.

G. Sales Concentration in Key States

State-level analysis identified California, New York, and Texas as the strongest revenue-generating markets.

These states collectively contribute a substantial portion of total sales and therefore represent strategically important territories for Nassau Candy Distributor.

Focused marketing initiatives and customer retention programs within these regions could further strengthen revenue performance.

X. RECOMENDATIONS

Based on the analysis findings, several strategic recommendations are proposed to improve profitability, optimize product performance, and support long-term business growth.

A. Prioritize High-Margin Products

Management should allocate greater promotional and operational resources toward products that generate both high sales and high profit margins.

Products within the top profitability rankings should receive priority in:

1. Marketing campaigns
2. Inventory planning
3. Demand forecasting
4. Distribution support

Protecting these products is essential because they represent the largest contributors to company profitability.

B. Review Low-Margin Products

Products exhibiting relatively low margins should be investigated to identify the root causes of reduced profitability.

Potential corrective actions include:

1. Cost reduction initiatives
2. Supplier renegotiation
3. Manufacturing optimization
4. Product repricing

Where improvements are not financially viable, management should consider product rationalization or discontinuation.

C. Reduce Product Portfolio Dependency

The analysis indicates a strong dependence on a limited number of chocolate products.

While these products currently perform exceptionally well, excessive concentration creates business risk.

Management should explore opportunities to strengthen:

- a. Sugar product performance
- b. Other product categories
- c. Emerging product lines

Portfolio diversification can improve long-term stability and reduce dependence on individual products.

D. Expand High-Profit Geographic Markets

Regions and states demonstrating superior profitability should receive additional strategic focus.

Examples include:

- a. Pacific Region
- b. California
- c. New York
- d. Texas

Targeted marketing investments and customer acquisition initiatives in these markets may generate stronger returns than expansion into lower-performing regions.

E. Implement Continuous Margin Monitoring

Profitability analysis should become a recurring business process rather than a one-time exercise.

Management should regularly monitor:

1. Gross Margin %
2. Profit Contribution
3. Revenue Contribution
4. Cost-to-Sales Ratios

The developed Streamlit dashboard provides an effective platform for real-time monitoring and decision support.

F. Use Profit-Based Performance Metrics

Future business decisions should prioritize profit-oriented indicators rather than relying exclusively on revenue metrics.

Performance evaluation frameworks should incorporate:

1. Gross Profit
2. Profit per Unit
3. Margin Percentage
4. Profit Contribution

This approach ensures that business growth translates into sustainable financial improvement.

XI. DASHBOARD STREAMLIT

To make the profitability analysis accessible to business stakeholders, an interactive web-based dashboard was developed using the Streamlit framework. The dashboard transforms raw transactional data into a dynamic decision-support system, allowing users to explore product performance, profitability trends, cost structures, and regional sales insights in real time.

The application was designed with simplicity and usability in mind, enabling managers and analysts to evaluate business performance without requiring technical expertise in data analytics or programming.

A. Dashboard Objectives

The primary objectives of the Streamlit dashboard are:

- Provide real-time visibility into product profitability.
- Monitor gross margin performance across divisions.
- Identify high-performing and underperforming products.
- Analyze relationships between sales, cost, profit, and units sold.
- Support strategic pricing and product portfolio decisions.
- Enable interactive exploration through filters and search capabilities.

B. Dashboard Modules

The dashboard consists of several analytical modules that align with the project's business objectives.

1) Product Profitability Overview

This module provides a comprehensive view of product-level performance.

Key features include:

- Top products by sales
- Top products by gross profit
- Product-level profitability rankings
- Revenue contribution analysis
- Profit contribution analysis

The module enables management to quickly identify products that generate the greatest financial value and distinguish them from products that generate revenue without contributing significantly to profitability.

2) Division Performance Dashboard

This module evaluates financial performance across product divisions.

Key visualizations include:

- Sales by division
- Average margin by division
- Revenue versus profit comparisons
- Division-level profitability indicators

The analysis revealed that the Chocolate Division consistently outperformed the Sugar and Other divisions in both revenue generation and margin efficiency.

3) Cost and Margin Diagnostics

This section focuses on understanding the relationship between sales, cost, and profitability.

Key visualizations include:

- ↳ Cost vs Sales scatter plot
- ↳ Sales vs Cost analysis
- ↳ Sales vs Profit analysis
- ↳ Units vs Profit analysis

These visualizations help identify products with excessive manufacturing costs, margin inefficiencies, or unusually low profitability despite strong sales performance.

4) Geographic Performance Analysis

This module evaluates profitability and revenue distribution across different regions and states.

Key visualizations include:

- ↳ Profit by Region
- ↳ Top States by Sales

The analysis supports market-focused decision-making

by identifying geographic areas that contribute the highest financial returns.

5) Trend Analysis Module

Time-series visualizations were incorporated to monitor changes in sales performance over time.

Key visualization:

Sales Over Time

This module helps identify seasonal patterns, growth trends, and periods of unusually high or low business activity.

C. Interactive Features

To enhance usability and analytical flexibility, the dashboard includes several interactive controls:

- Date Range Selection
- Division Filter
- Product Search
- Margin Threshold Filtering
- Dynamic Chart Updates

These features allow users to customize their analysis according to specific business requirements and investigate profitability patterns from multiple perspectives.

D. Business Value of the Dashboard

The developed Streamlit application serves as a practical business intelligence tool that converts transactional data into actionable insights.

The dashboard enables decision-makers to:

- ↳ Monitor profitability in real time.
- ↳ Identify margin risks early.
- ↳ Improve pricing strategies.
- ↳ Optimize product portfolios.
- ↳ Support data-driven strategic planning.

By integrating financial metrics, visual analytics, and interactive exploration capabilities, the dashboard significantly improves the organization's ability to make informed profitability-focused decisions.

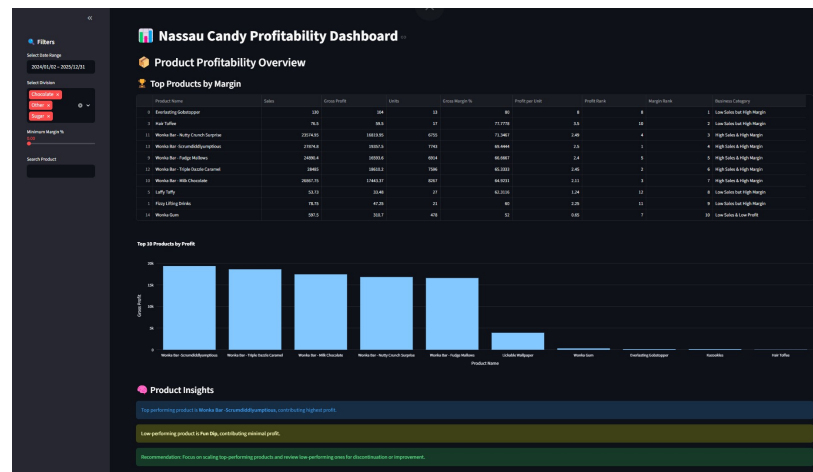


Fig. 14. Dashboard Streamlit

XII. FUTURE SCOPE

The current system provides detailed insights into product profitability, margin performance, divisional efficiency, and regional sales trends. Future enhancements may include the integration of predictive analytics and forecasting models to estimate future sales, costs, and profitability, enabling more proactive decision-making. The Streamlit dashboard can also be connected with real-time sales, inventory, and ERP systems for continuous business monitoring. Additional analysis of factory-level operations, supply chain performance, and transportation efficiency could further improve operational effectiveness. If customer demographic and behavioral data become available, customer segmentation techniques such as K-Means clustering may be applied to identify purchasing patterns and support targeted marketing strategies. These enhancements would transform the current profitability analysis system into a comprehensive predictive business intelligence platform that supports long-term strategic planning and profitability optimization.

XIII. CONCLUSION

This study developed a comprehensive Product Line Profitability and Margin Performance Analysis framework for Nassau Candy Distributor using transactional sales, cost, profit, and product-level data.

The analysis demonstrated that profitability is highly concentrated among a relatively small number of products, particularly within the Chocolate Division. While these products generate significant revenue and profit, the findings also revealed the importance of evaluating margins and cost structures alongside sales performance. Division-level analysis identified Chocolate as the strongest performing business segment, while regional and state-level evaluations highlighted important geographic variations in profitability. Correlation analysis further confirmed the strong relationship between sales growth and profit generation.

Unlike customer segmentation studies, this project did not require machine learning clustering techniques such as K-Means because the primary objective was financial performance evaluation rather than customer grouping. Instead, profitability metrics, aggregation analysis, margin diagnostics, correlation analysis, and business intelligence visualizations provided more meaningful and actionable insights.

The developed Streamlit dashboard transforms raw transactional data into an interactive decision-support system that enables managers to evaluate product profitability, monitor margin performance, identify cost inefficiencies, and make data-driven strategic decisions.

Overall, the project provides Nassau Candy Distributor with a practical framework for improving profitability, optimizing product portfolios, and supporting sustain-

able business growth through evidence-based financial analysis.